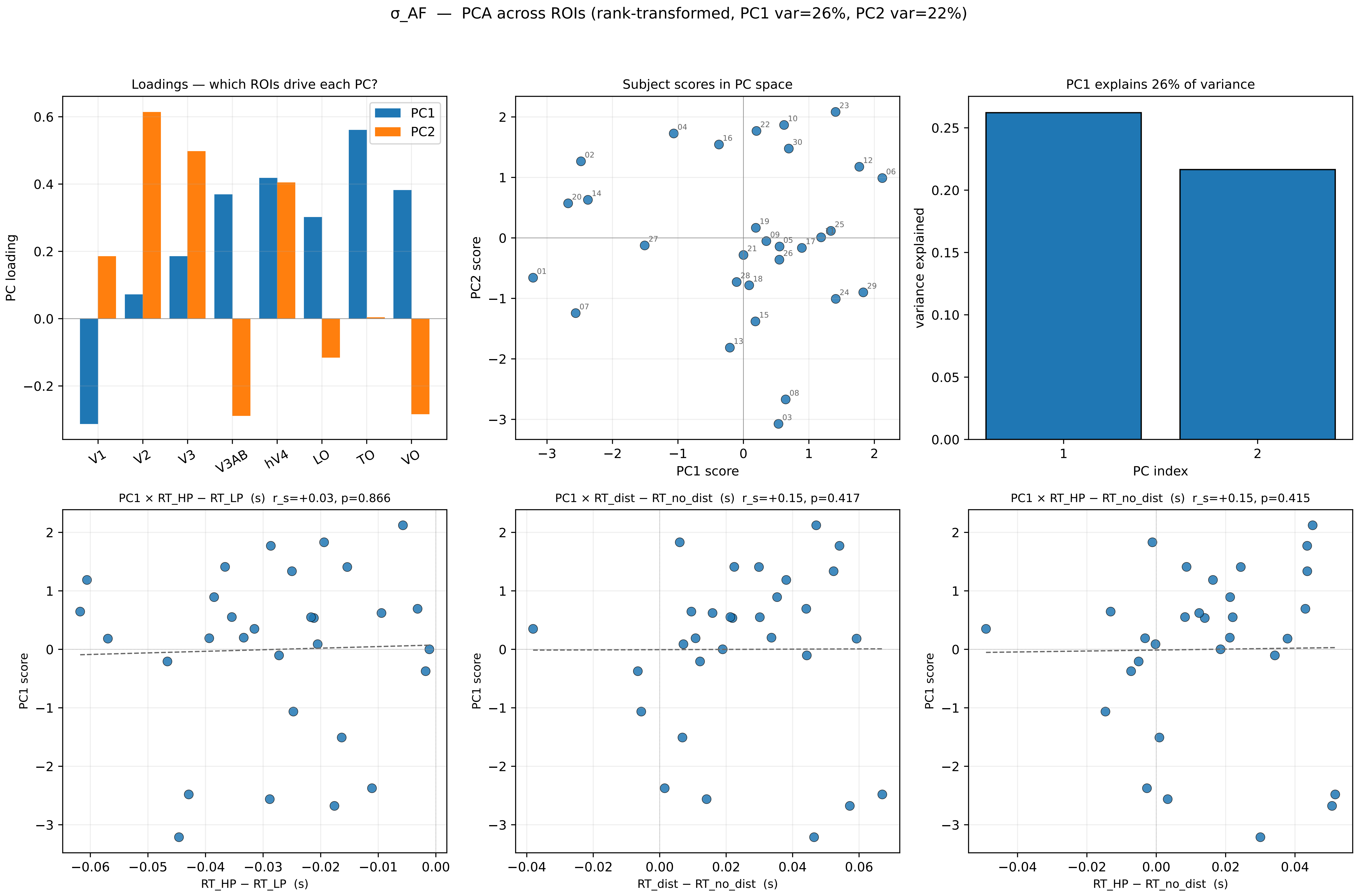
Per-parameter PCA across ROIs × behavior

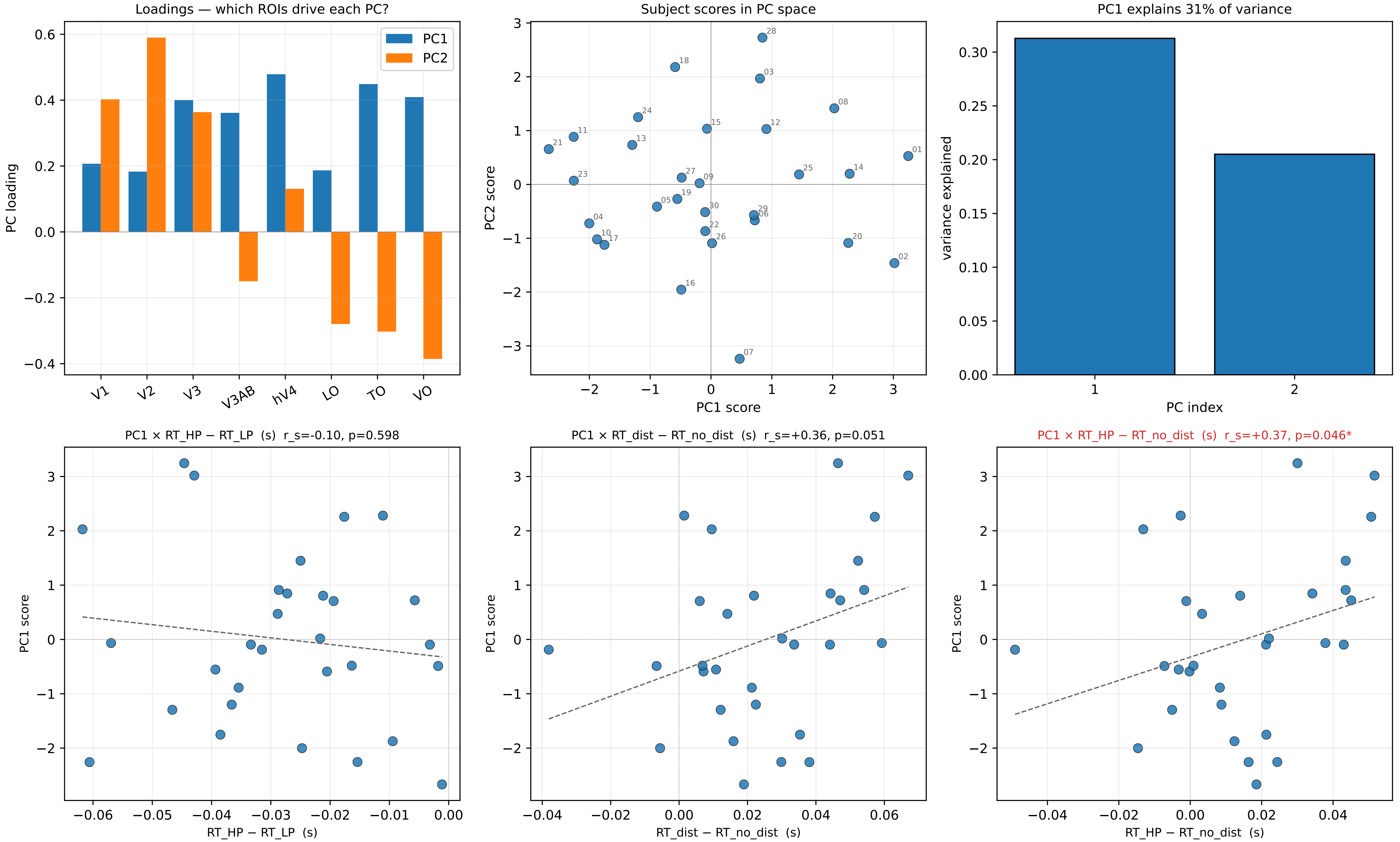
- For each AF parameter, build a (subject × ROI) matrix (rank-transformed per ROI, then standardized).
- Run PCA. PC1 captures the largest shared subject-level variance across the visual hierarchy.
- Correlate per-subject PC1 with each behavioral contrast.

Why this is more powerful than per-ROI:

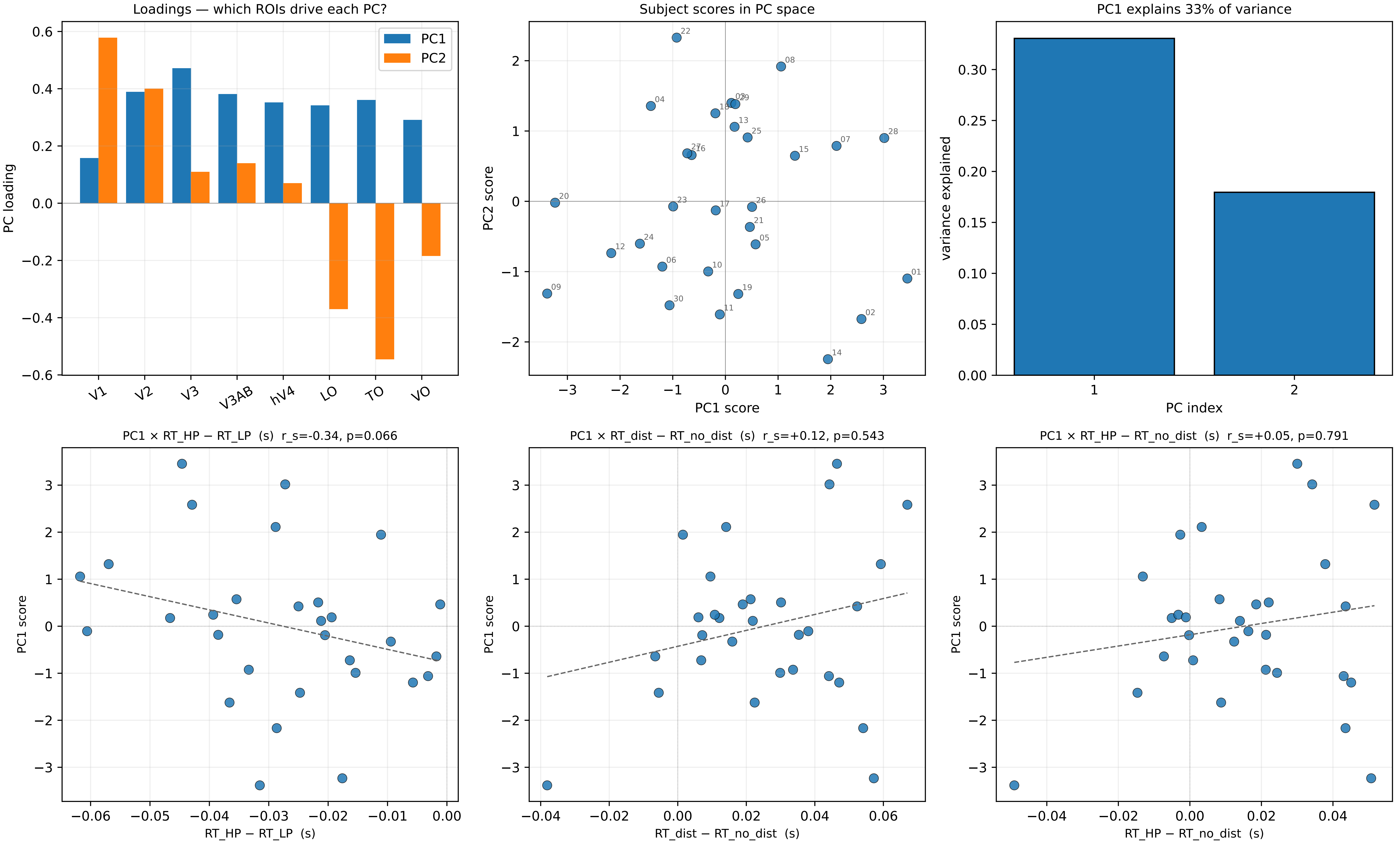
Per-ROI behavior correlations test 8 weak signals independently. PC1 pools the SIGNAL across ROIs and rejects ROI-specific noise — one strong test per parameter instead of 8 weak ones.

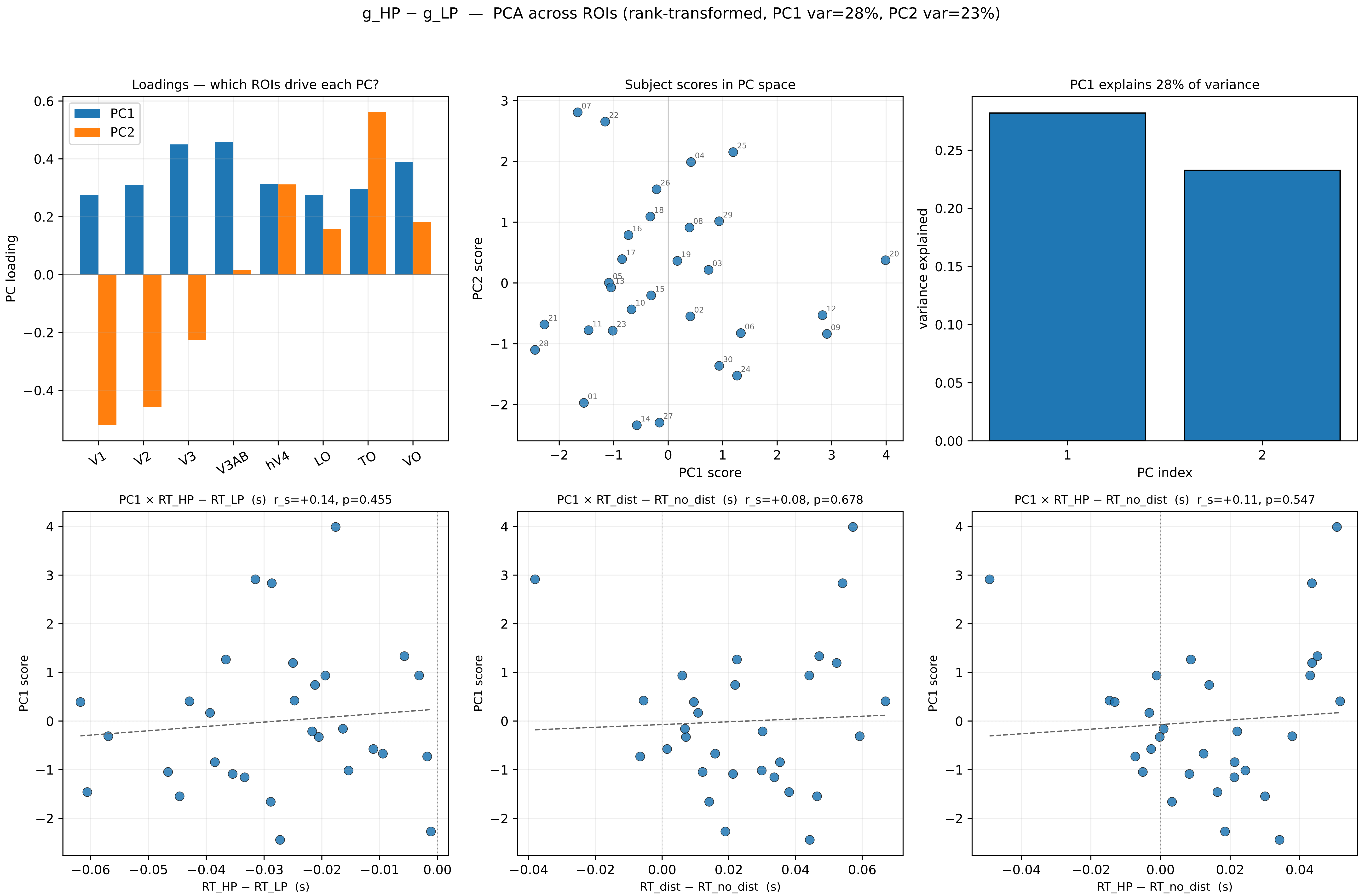

σ_AF — PCA across ROIs (rank-transformed, PC1 var=26%, PC2 var=22%)
Loadings — which ROIs drive each PC?
PC1
PC2
PC loading
V1
V2
V3
V3AB
hV4
LO
TO
VO
Subject scores in PC space
PC1 score
PC2 score
PC1 explains 26% of variance
variance explained
PC index
PC1 × RT_HP − RT_LP (s) r_s=+0.03, p=0.866
RT_HP − RT_LP (s)
PC1 × RT_dist − RT_no_dist (s) r_s=+0.15, p=0.417
RT_dist − RT_no_dist (s)
PC1 × RT_HP − RT_no_dist (s) r_s=+0.15, p=0.415
RT_HP − RT_no_dist (s)

g_HP — PCA across ROIs (rank-transformed, PC1 var=31%, PC2 var=21%)



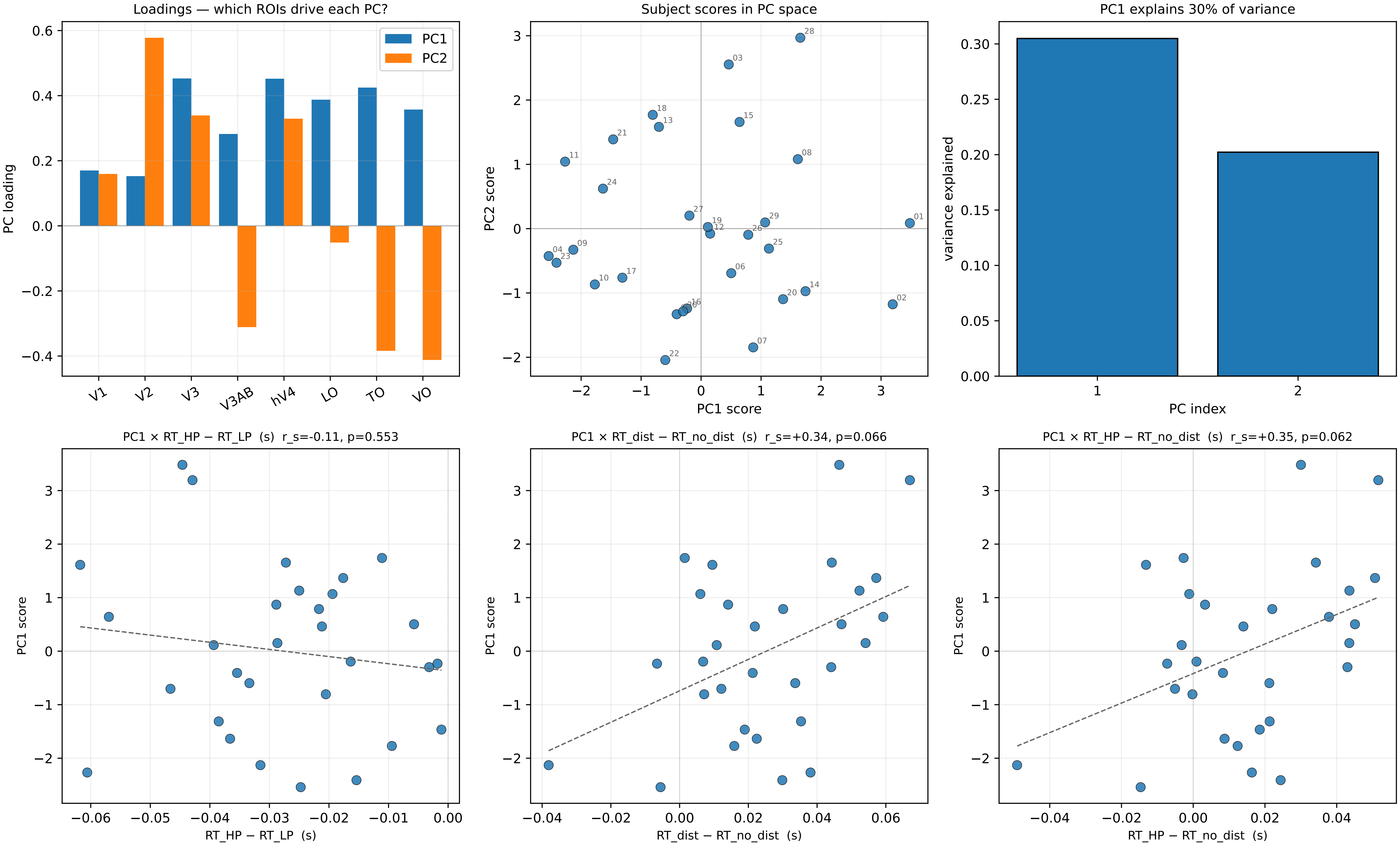
g_LP — PCA across ROIs (rank-transformed, PC1 var=33%, PC2 var=18%)





g_HP − g_LP — PCA across ROIs (rank-transformed, PC1 var=28%, PC2 var=23%)
Loadings — which ROIs drive each PC?
PC1
PC2
PC loading
V1
V2
V3
V3AB
hV4
LO
TO
VO
Subject scores in PC space
PC1 score
PC2 score
PC1 explains 28% of variance
variance explained
PC index
PC1 × RT_HP − RT_LP (s) r_s=+0.14, p=0.455
RT_HP − RT_LP (s)
PC1 score
PC1 × RT_dist − RT_no_dist (s) r_s=+0.08, p=0.678
RT_dist − RT_no_dist (s)
PC1 score
PC1 × RT_HP − RT_no_dist (s) r_s=+0.11, p=0.547
RT_HP − RT_no_dist (s)
PC1 score

½(g_HP + g_LP) — PCA across ROIs (rank-transformed, PC1 var=30%, PC2 var=20%)



log((1+g_HP)/(1+g_LP)) — PCA across ROIs (rank-transformed, PC1 var=29%, PC2 var=22%)